

STUDY QUESTIONS 3 - ANSWERS

1. A warehouse manager-

a) Decision variables:

Let $J = \{A, B, C, D, \dots\}$ be the set of all potential products to stock.

$$x_j = \begin{cases} 1 & \text{if product } j \in J \text{ is stocked in the warehouse} \\ 0 & \text{otherwise} \end{cases} \quad \forall j \in J$$

b) Complete Mathematical Formulation:

p_j : Profit of product j

w_j : Weight of product j

s_j : Space occupied by product j

S : Available floor space (space limitation) of the warehouse

W : A maximum load-bearing capacity for the shelving system

Objective Function (maximize total profit):

$$\max \sum_{j \in J} p_j \cdot x_j$$

Subject to:

Floor space constraint:

$$\sum_{j \in J} s_j \cdot x_j \leq S$$

Weight capacity constraint:

$$\sum_{j \in J} w_j \cdot x_j \leq W$$

Variable type constraints:

$$x_j \in \{0, 1\} \quad \forall j \in J$$

c) Specific Model for Given Products:

Objective Function:

$$\max \quad 800x_A + 650x_B + 900x_C + 550x_D$$

Subject to:

$$12x_A + 8x_B + 15x_C + 6x_D \leq 30$$

$$150x_A + 200x_B + 120x_C + 180x_D \leq 400$$

$$x_j \in \{0, 1\} \quad \forall j \in \{A, B, C, D\}$$

2. Factory Production Planning Problem-

a) Decision variables:

x : Number of cars produced per month.

y : Number of bicycles produced per month.

b) **Complete Mathematical Formulation:**

Objective Function (maximize profit):

$$\text{maximize } Z = 2500x + 120y$$

Subject to:

Steel Constraint: A car uses 1500 grams of steel (1.5 kg), and a bicycle uses 20 kg of steel. Our maximum available steel is 30000 kg.

$$1.5x + 20y \leq 30,000$$

Labor Constraint: A car needs 30 hours of labor, and a bicycle needs 2 hours of labor. Our total available labor is 2,400 hours.

$$30x + 2y \leq 2,400$$

Assembly Time Constraint: A car takes 20 hours of assembly, and a bicycle takes 3 hours of assembly. The total available assembly time is 1,800 hours.

$$20x + 3y \leq 1,800$$

Market Demand Constraint: The factory cannot sell more than 15 cars and the factory must produce at least 50 bicycles.

$$x \leq 15$$

$$y \geq 50$$

Variable type constraints: Non-negativity

$$x \geq 0$$

$$y \geq 0$$

3. **Production Scheduling in a Sweet Factory-**

a) **Decision variables:**

Let $I = \{1, 2\}$ be the set of months in which production will take place.

x_i : Number of chocolate boxes produced in month $i \in I$.

y_i : Number of cookie boxes produced in month $i \in I$.

b) **Complete Mathematical Formulation:**

Objective Function (minimize the total production time):

$$\text{minimize } \sum_{i \in I} 4x_i + \sum_{i \in I} 2y_i$$

Subject to:

Monthly Production Capacity Constraints:

$$4x_i + 2y_i \leq 600 \quad \forall i \in I$$

Demand Constraints:

At least 100 boxes of cookies must be produced by the end of month 1.

$$y_1 \geq 100$$

At least 150 boxes of chocolates must be produced by the end of month 2.

$$\sum_{i \in I} x_i \geq 150$$

Variable type constraints: Non-negativity

$$x_i \geq 0 \quad \forall i \in I$$

$$y_i \geq 0 \quad \forall i \in I$$

c) **Complete Mathematical Formulation: Open Form based on given data**

$$\begin{aligned}
 &\text{minimize } z = 4x_1 + 4x_2 + 2y_1 + 2y_2 \\
 &\text{subject to} \\
 &\quad 4x_1 + 2y_1 \leq 600 \\
 &\quad 4x_2 + 2y_2 \leq 600 \\
 &\quad y_1 \geq 100 \\
 &\quad x_1 + x_2 \geq 150 \\
 &\quad x_1, x_2, y_1, y_2 \geq 0
 \end{aligned}$$

4. **Preparing a Healthy Feed Mix-**

a) **Decision variables:**

x : Kilograms of corn used per day

y : Kilograms of soybean meal used per day

z : Kilograms of hay used per day

b) **Complete Mathematical Formulation:**

Objective Function (minimize total daily cost of the feed):

$$\text{minimize } Z = 0.5x + 0.8y + 0.3z$$

Subject to:

Protein Constraint: Corn provides 4 units/kg, Soybean Meal provides 10 units/kg and Hay provides 2 units/kg. Minimum requirement is 50 units of protein.

$$4x + 10y + 2z \geq 50$$

Fiber Constraint: Corn provides 2 units/kg, Soybean Meal provides 1 unit/kg and Hay provides 5 units/kg. Minimum requirement is 30 units of fiber.

$$2x + y + 5z \geq 30$$

Energy Constraint: Corn provides 20 units/kg, Soybean Meal provides 15 units/kg and Hay provides 10 units/kg. Minimum requirement is 180 units of energy.

$$20x + 15y + 10z \geq 180$$

Variable type constraints: Non-negativity

$$x \geq 0$$

$$y \geq 0$$

$$z \geq 0$$

5. **Steel Rod Cutting-**

a) **Decision variables:**

Let $I = \{1, 2, 3, 4, 5\}$ be the patterns used when cutting rods.

x_i : Number of rods cut using pattern $i \in I$

b) **Complete Mathematical Formulation:**

Objective Function (maximize total profit):

p_i : Profit of pattern i

$$p_1 = 5 * 4 = 20 \quad \rightarrow 5 \times 20 \text{ inch}$$

$$p_2 = 3 * 6 = 18 \quad \rightarrow 3 \times 30 \text{ inch}$$

$$p_3 = 2 * 10 = 20 \quad \rightarrow 2 \times 50 \text{ inch}$$

$$p_4 = 1 * 10 + 1 * 4 + 1 * 6 = 10 + 4 + 6 = 20 \quad \rightarrow 1 \times 50 \text{ inch} + 1 \times 30 \text{ inch} + 1 \times 20 \text{ inch}$$

$$p_5 = 1 * 10 + 2 * 4 = 18 \quad \rightarrow 1 \times 50 \text{ inch} + 2 \times 20 \text{ inch}$$

$$\text{maximize } \sum_{i \in I} p_i x_i \quad OR \quad 20x_1 + 18x_2 + 20x_3 + 20x_4 + 18x_5$$

Subject to:

20-inch Rod Demand: Pattern 1 gives 5 pieces of rod, pattern 4 gives 1 piece of rod and pattern 5 gives 2 pieces of rod. Also 20-inch Rod is demanded as at least 80 pieces of rod.

$$5x_1 + x_4 + 2x_5 \geq 80$$

30-inch Rod Demand: Pattern 2 gives 3 pieces of rod and pattern 4 gives 1 piece of rod. Also 30-inch Rod is demanded as at least 60 pieces of rod.

$$3x_2 + x_4 \geq 60$$

50-inch Rod Demand: Pattern 3 gives 2 pieces of rod, pattern 4 gives 1 piece of rod and pattern 5 gives 1 piece of rod. Also 50-inch Rod is demanded as at least 40 pieces of rod.

$$2x_3 + x_4 + x_5 \geq 40$$

Variable type constraints: Non-negativity

$$x_i \geq 0 \quad \forall i \in I$$

6. Staff Assignment at a Summer Camp-

a) **Decision variables:**

Let $I = \{Alice, Ben, Chole, David, Emma, Frank\}$ be the set of staff.

And, $J = \{Swimming, Arts\&Crafts, Hiking\}$ be the set of activities.

$$x_{ij} = \begin{cases} 1 & \text{if staff } i \in I \text{ is assigned to activity } j \in J \\ 0 & \text{otherwise} \end{cases}$$

b) **Complete Mathematical Formulation:**

Objective Function (maximize total skill score of weekly assignment):

c_{ij} : Staff i 's skill score at activity j

$$\text{maximize } \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}$$

Subject to:

Time Constraint – Weekly Assignment Constraint: We can only assign a staff to one activity per week. Thus, among all decision variables of a certain staff one of them must be 1 and the rest is 0.

$$\sum_{j \in J} x_{ij} = 1 \quad \forall i \in I$$

Staffing Constraint: We will also need to staff a task exactly by 2 people. Thus, among all decision variables of a certain task two of them must be 1 and the rest is 0.

$$\sum_{i \in I} x_{ij} = 2 \quad \forall j \in J$$

Variable type constraints:

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$$

7. Assigning Technicians-

a) Decision variables:

Let $I = \{John, Maria, Kevin, Sophie\}$ be the set of technicians.

And, $J = \{JobA, JobB, JobC, JobD\}$ be the set of jobs.

$$x_{ij} = \begin{cases} 1 & \text{if technician } i \in I \text{ is assigned to job } j \in J \\ 0 & \text{otherwise} \end{cases}$$

b) Complete Mathematical Formulation:

Objective Function (minimize the total time required to complete all four jobs):

t_{ij} : Technician i 's estimated time to finish repairing job j

$$\text{minimize} \quad \sum_{i \in I} \sum_{j \in J} t_{ij} x_{ij}$$

Subject to:

Assignment Constraint: At most, we can assign a single task to a technician.

$$\sum_{j \in J} x_{ij} = 1 \quad \forall i \in I$$

Staffing Constraint: We must assign a single worker to a task.

$$\sum_{i \in I} x_{ij} = 1 \quad \forall j \in J$$

Variable type constraints:

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J$$

8. Laptop Transportation-

a) Decision variables:

Let $I = \{Chicago, Dallas\}$ be the set of technicians.

And, $J = \{NewYork, Atlanta, Denver\}$ be the set of jobs.

x_{ij} = Number of laptops shipped from factory i to warehouse j

b) Complete Mathematical Formulation:

Objective Function (minimize the cost of total transportation):

c_{ij} : Cost of transportation of a single laptop from factory i to warehouse j

$$\text{minimize} \quad \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}$$

Subject to:

Demand Satisfaction: The total amount shipped to a warehouse must satisfy the demand of that particular warehouse.

d_j : Number of laptops demanded by warehouse j

$$\sum_{i \in I} x_{ij} = d_j \quad \forall j \in J$$

Supply Constraint: Total amount shipped from a factory must not exceed the supply of that particular factory.

s_i : Supply of factory i

$$\sum_{j \in J} x_{ij} = s_i \quad \forall i \in I$$

Variable type constraints:

$$\begin{aligned} x_{ij} &\geq 0 \quad \forall i \in I, j \in J \\ x_{ij} &\in Z \quad \forall i \in I, j \in J \end{aligned}$$

9. Emma's Investment Portfolio-

a) Decision variables:

Let $I = \{GovernmentBonds, RealEstateFund, TechStocks, InternationalIndexFund, High - RiskStart - upPortfolio\}$ be the set of investment options.

x_i = Invested amounts (\$) via investment option i

b) Complete Mathematical Formulation:

Objective Function (maximize the total return of the investment):

r_i : The expected annual return (%) for option i

$$\text{maximize} \quad \sum_{i \in I} r_i x_i \quad OR \quad 0.04x_1 + 0.06x_2 + 0.1x_3 + 0.07x_4 + 0.15x_5$$

Subject to:

Investment Limit Constraints: We cannot invest more than 400,000 dollars in Tech Stocks or Start-up Portfolio combined. Additionally, we cannot invest less than 200,000 dollars to Government Bonds.

$$\begin{aligned} x_1 &\geq 200,000 \\ x_3 + x_5 &\leq 400,000 \end{aligned}$$

Budget Allocation Constraint: All budgets must be allocated. There must not be any leftover from the initial budget.

$$\sum_{i \in I} x_i = 1,000,000 \quad OR \quad x_1 + x_2 + x_3 + x_4 + x_5 = 1,000,000$$

Variable type constraints:

$$\begin{aligned} x_i &\geq 0 \quad \forall i \in I \\ x_i &\in R \quad \forall i \in I \end{aligned}$$

10. CPU Production in AI Race-

a) Decision variables:

X_A = Number of Type A CPU Units Produced.

X_B = Number of Type B CPU Units Produced.

X_C = Number of Type C CPU Units Produced.

b) Complete Mathematical Formulation:

Objective Function (maximize total revenue):

$$\text{maximize} \quad 9500X_A + 5500X_B + 7000X_C$$

Subject to:

Resource Constraints:

$$4X_A + 2X_B + 3X_C \leq 4500$$

$$5X_A + 8X_B + 6X_C \leq 8000$$

$$4X_A + 3X_B + 6X_C \leq 5000$$

Sustainability Constraints:

$$160X_A + 80X_B + 115X_C \leq 350.000$$

$$45X_A + 25X_B + 30X_C \leq 65.000$$

Variable type constraints:

$$X_A, X_B, X_C \geq 0$$

$$X_A, X_B, X_C \in Z$$

11. EBG Logistics' Transportation Problem-

a) **Decision variables:**

x_{ij} = Amount of items transported from warehouse W_i to store S_j (units).

b) **Complete Mathematical Formulation:**

Objective Function (minimize transportation cost):

We pay a transportation fee per item additionally we also pay customs fee from items imported from warehouses W_3 and W_4 :

c_{ij} = Cost of transportation per item from warehouse W_i to store S_j (\$ / units).

$$\text{minimize} \quad \sum_{i=1}^5 \sum_{j=1}^5 x_{ij} c_{ij} + \sum_{j=1}^5 8x_{3j} + \sum_{j=1}^5 8x_{4j}$$

Subject to:

t_i = Available amount of items in warehouse W_i .

d_j = Demand of store S_j .

Storage Constraint:

$$\sum_{j=1}^5 x_{ij} \leq t_i \quad \forall i \in \{1, 2, 3, 4, 5\}$$

Demand Satisfaction Constraint:

$$\sum_{i=1}^5 x_{ij} = d_j \quad \forall j \in \{1, 2, 3, 4, 5\}$$

Variable type constraints:

$$x_{ij} \geq 0 \quad \forall i \in \{1, 2, 3, 4, 5\}, j \in \{1, 2, 3, 4, 5\}$$

$$x_{ij} \in Z$$